

```
import numpy as numpy
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
df=pd.read_csv("/content/Bike Sharing Demand.csv")
```

```
df.head()
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count	
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0	3	13	16	
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0	8	32	40	
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0	5	27	32	
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0	3	10	13	
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0	0	1	1	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.tail
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count	
10881	2012-12-19 19:00:00	4	0	1	1	15.58	19.695	50	26.0027	7	329	336	
10882	2012-12-19 20:00:00	4	0	1	1	14.76	17.425	57	15.0013	10	231	241	
10883	2012-12-19 21:00:00	4	0	1	1	13.94	15.910	61	15.0013	4	164	168	
10884	2012-12-19 22:00:00	4	0	1	1	13.94	17.425	61	6.0032	12	117	129	
10885	2012-12-19 23:00:00	4	0	1	1	13.12	16.665	66	8.9981	4	84	88	

```
df.shape
```

```
(10886, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10886 entries, 0 to 10885
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   datetime    10886 non-null  object
1   season      10886 non-null  int64
2   holiday     10886 non-null  int64
3   workingday  10886 non-null  int64
4   weather     10886 non-null  int64
5   temp        10886 non-null  float64
6   atemp       10886 non-null  float64
7   humidity    10886 non-null  int64
8   windspeed   10886 non-null  float64
9   casual      10886 non-null  int64
10  registered  10886 non-null  int64
11  count       10886 non-null  int64
```

```
dtypes: float64(3), int64(8), object(1)
memory usage: 1020.7+ KB
```

```
df.isnull().sum()
```

	0
datetime	0
season	0
holiday	0
workingday	0
weather	0
temp	0
atemp	0
humidity	0
windspeed	0
casual	0
registered	0
count	0

```
dtype: int64
```

```
df
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count	
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0000	3	13	16	
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0000	8	32	40	
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0000	5	27	32	
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0000	3	10	13	
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0000	0	1	1	
...	...	...	...	...	...	...	...	...	...	...	...	...	
10881	2012-12-19 19:00:00	4	0	1	1	15.58	19.695	50	26.0027	7	329	336	
10882	2012-12-19 20:00:00	4	0	1	1	14.76	17.425	57	15.0013	10	231	241	
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10885	2012-12-19 23:00:00	4	0	1	1	13.12	16.665	66	8.9981	4	84	88	

10886 rows × 12 columns

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df = pd.read_csv("/content/Bike Sharing Demand.csv")
x = df[['season', 'holiday', 'workingday', 'weather', 'temp', 'atemp', 'humidity', 'windspeed']]
y = df['count']
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
model=LinearRegression()
model.fit(x_train,y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
y_pred=model.predict(x_test)
mse=mean_squared_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(mse)
print(r2)
```

```
23906.814693176104
0.2757033009386365
```

```
new = pd.DataFrame({
    'season': [1], # Example value: winter
    'holiday': [0], # Example value: not a holiday
    'workingday': [1], # Example value: working day
    'weather': [1], # Example value: clear weather
    'temp': [9.89], # The temperature you provided
    'atemp': [14.0], # Example value for felt temperature
    'humidity': [60], # Example value for humidity (adjusted)
    'windspeed': [10.0] # Example value for windspeed (in m/s)
})
predicted_count = model.predict(new)
print("predicted_count", predicted_count[0])
```

```
predicted_count 86.44332030707312
```